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Java SE 27 & JDK 27

Interface NetworkChannel

All Superinterfaces:

AutoCloseable, Channel, Closeable

All Known Subinterfaces:

MulticastChannel

All Known Implementing Classes:

AsynchronousServerSocketChannel, AsynchronousSocketChannel, DatagramChannel, ServerSocketChannel, SocketChannel

---

```
public interface NetworkChannel
extends Channel
```

A channel to a network socket.

A channel that implements this interface is a channel to a network socket. The `bind` method is used to bind the socket to a local `address`, the `getLocalAddress` method returns the address that the socket is bound to, and the `setOption` and `getOption` methods are used to set and query socket options. An implementation of this interface should specify the socket options that it supports.

The `bind` and `setOption` methods that do not otherwise have a value to return are specified to return the network channel upon which they are invoked. This allows method invocations to be chained. Implementations of this interface should specialize the return type so that method invocations on the implementation class can be chained.

Since:

1.7

Method Summary

All Methods	Instance Methods	
Modifier and Type	**Method**	**Description**
NetworkChannel	**bind** (SocketAddress local)	Binds the channel's socket to a local address.
SocketAddress	**getLocalAddress**()	Returns the socket address that this channel's socket is bound to.
<T> T		

	getOption(SocketOption<T> name)	Returns the value of a socket option.
<T> NetworkChannel	**setOption**(SocketOption<T> name, T value)	Sets the value of a socket option.
Set<SocketOption<?>>	**supportedOptions**()	Returns a set of the socket options supported by this channel.

Methods declared in interface Channel

close, isOpen

Method Details

bind

```
NetworkChannel bind(SocketAddress local)
             throws IOException
```

Binds the channel's socket to a local address.

This method is used to establish an association between the socket and a local address. Once an association is established then the socket remains bound until the channel is closed. If the `local` parameter has the value `null` then the socket will be bound to an address that is assigned automatically.

Parameters:

`local` - The address to bind the socket, or `null` to bind the socket to an automatically assigned socket address

Returns:

This channel

Throws:

`AlreadyBoundException` - If the socket is already bound

`UnsupportedAddressTypeException` - If the type of the given address is not supported

`ClosedChannelException` - If the channel is closed

`IOException` - If some other I/O error occurs

See Also:

`getLocalAddress()`

getLocalAddress

```
SocketAddress getLocalAddress()
                        throws IOException
```

Returns the socket address that this channel's socket is bound to.

Where the channel is bound to an Internet Protocol socket address then the return value from this method is of type InetSocketAddress.

Returns:

The socket address that the socket is bound to, or null if the channel's socket is not bound

Throws:

ClosedChannelException - If the channel is closed

IOException - If an I/O error occurs

setOption

```
<T> NetworkChannel setOption(SocketOption<T> name,
                             T value)
                      throws IOException
```

Sets the value of a socket option.

Type Parameters:

T - The type of the socket option value

Parameters:

name - The socket option

value - The value of the socket option. A value of null may be a valid value for some socket options.

Returns:

This channel

Throws:

UnsupportedOperationException - If the socket option is not supported by this channel

IllegalArgumentException - If the value is not a valid value for this socket option

ClosedChannelException - If this channel is closed

IOException - If an I/O error occurs

See Also:

StandardSocketOptions

getOption

```
<T> T getOption(SocketOption<T> name)
         throws IOException
```

Returns the value of a socket option.

Type Parameters:

T - The type of the socket option value

Parameters:

name - The socket option

Returns:

The value of the socket option. A value of `null` may be a valid value for some socket options.

Throws:

UnsupportedOperationException - If the socket option is not supported by this channel

ClosedChannelException - If this channel is closed

IOException - If an I/O error occurs

See Also:

StandardSocketOptions

supportedOptions

```
Set<SocketOption<?>> supportedOptions()
```

Returns a set of the socket options supported by this channel.

This method will continue to return the set of options even after the channel has been closed.

Returns:

A set of the socket options supported by this channel

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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